📖 Computer Graphics 😎 ⚡🤭 Note 🤫⚡

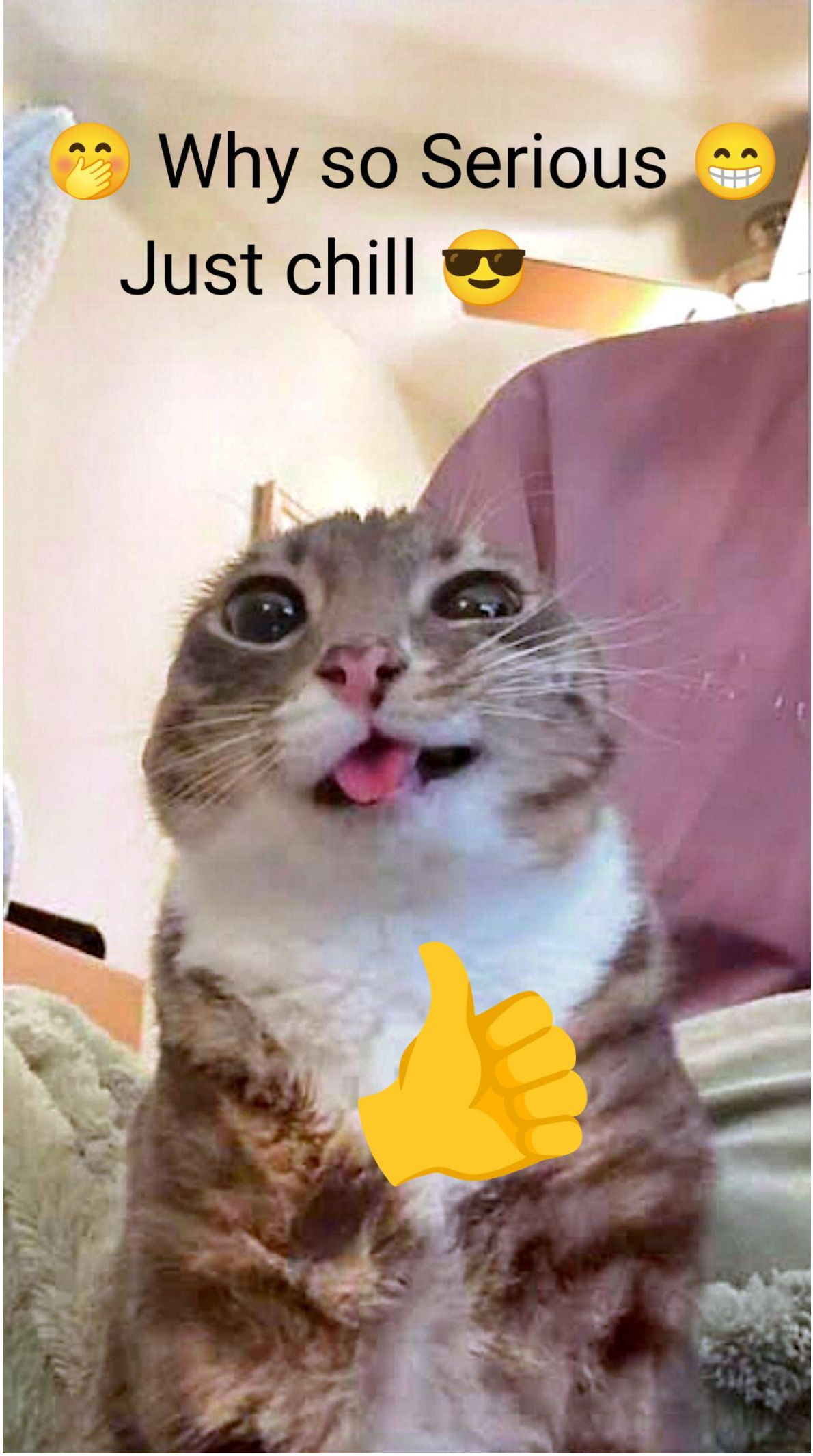
👉 Mid Term (2026) 👈

🔥 Topic 👉 01 : DDA



Created By 😎
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Computer Graphics

Mid term

MCQ ⟶ 20 × 1 o 5 = 30 mark

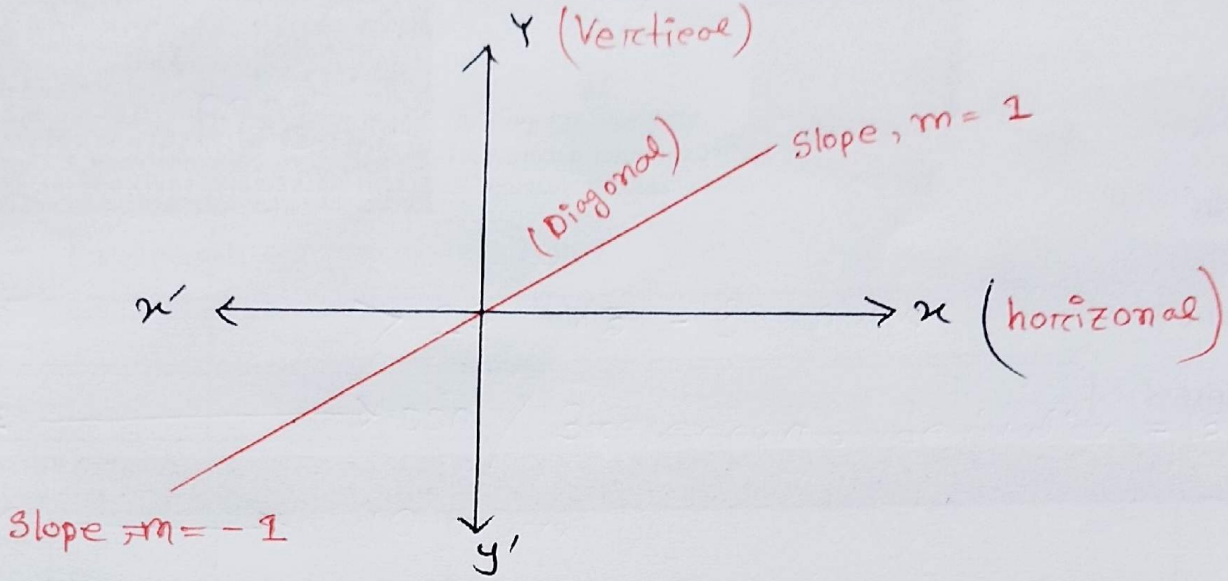
Obj ⟶ DDA / Bresenham's derivation = 15 mark

code ⟶ 10 mark

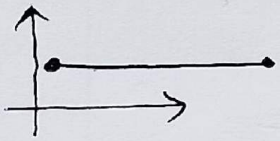
theory ⟶ 45 mark (Ans 3 out of 4)

So,
- Slide-1 theory
- RGB, CMYK math + theory
- DDA
- Bresenham's

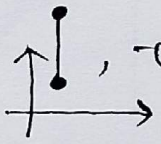
Incremental Algorithm



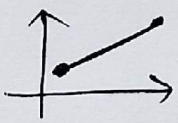
আমরা জানি,

* horizontal line,  অর্থাৎ, $x_{n+1} = x_n + 1$,
$y_{n+1} = y_n$

অর্থাৎ x বৃদ্ধি পাবে, x increment by 1,

* Vertical line,  অর্থাৎ, $x_{n+1} = x_n$, $y_{n+1} = y_n + 1$

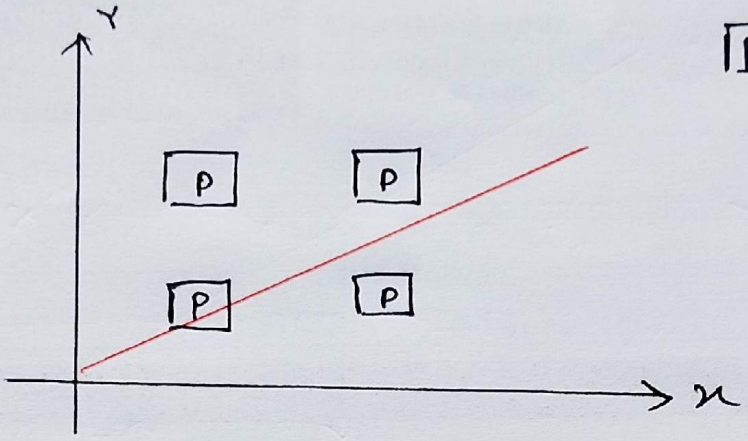
অর্থাৎ y বৃদ্ধি পাবে, y increment by 1,

* Digonal line:  এখন, $x_{n+1} = x_n + 1$
$y_{n+1} = y_n + 1$

Both x, y increment by 1,

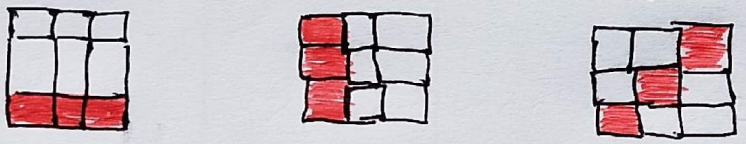
এখন এদের বাইরে line এর জন্য আমরা Algorithum ব্যবহার করে নিতে পারি।

Topic – 01 : DDA (Digital Differential Analyzer)

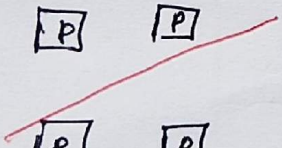


আমরা জানি, pixel মানে ওই ছোট unit স্কোয়ার colour হয়, আমরা visual দেখতে পাই,

যেমন, horizontal, vertical, diagonal



So যদি আরো কাছে থেকে দেখি, তবে ঠিক উপরের Graph এর মতো ফাঁকা থাকে।



তখন line এর খুব কাছে pixel গুলোই select হয়।

DDA তে সমস্যা, calculation Floating point আসে যা Round up করে pixel select হয়, 4.5 = 5, 4.49 = 4

DDA formula :

যদি, $-1 < m < 1 \Rightarrow$

$$x_{new} = x_{old} + 1$$

$$y_{new} = y_{old} + m$$

$m = 1$,

$x = x + 1$

$y = y + 1$

$m > 1 \Rightarrow$

$$x_{new} = x_{old} + \frac{1}{m}$$

$$y_{new} = y_{old} + 1$$

$m = -1$,

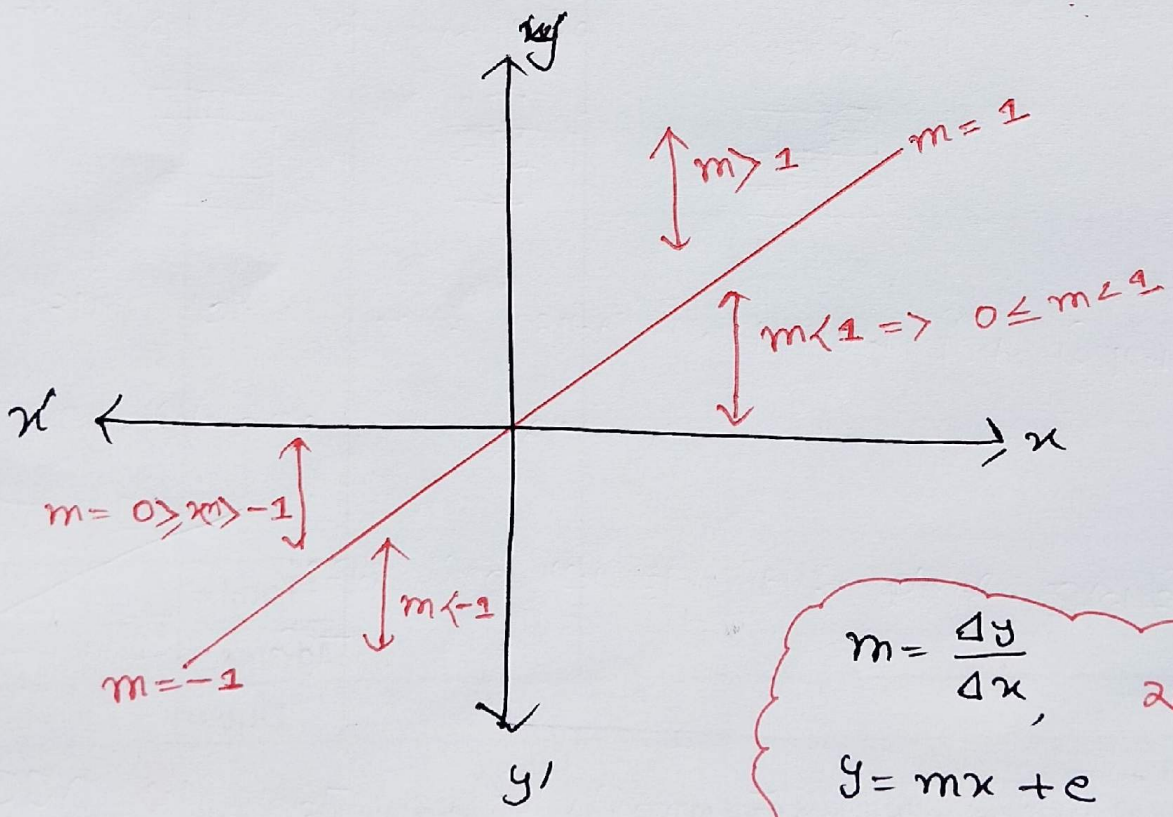
$x = x - 1$

$y = y - 1$

$m < -1 \Rightarrow$

$$x_{new} = x_{old} - \frac{1}{m}$$

$$y_{new} = y_{old} - 1$$



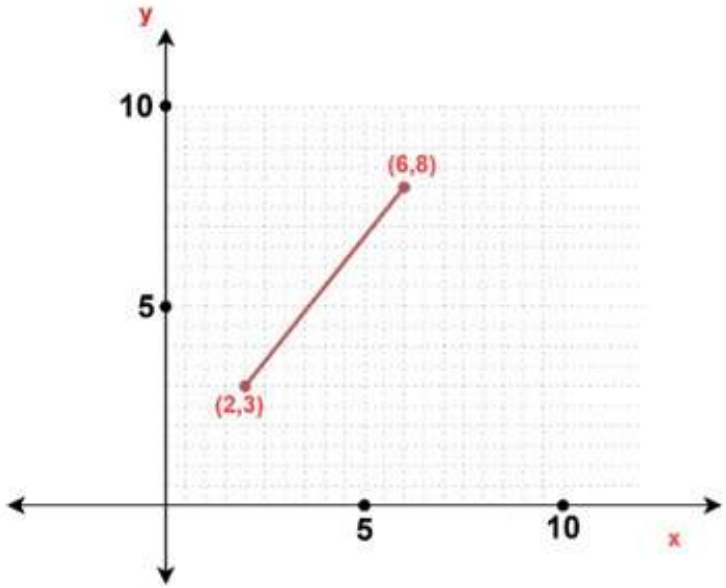
$m = \frac{\Delta y}{\Delta x}$, এবং

$y = mx + c$ রাখলে

$ax + bx + c = 0$

Practice Problem

Problem1: Use the DDA algorithm to determine the pixel positions for the given line segment from (x_1, y_1) to (x_2, y_2) as shown in the graph.



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: practice problem of DDA :

Ans to the Question no 1 :

Given, $x_0 = 2$, $y_0 = 3$

$x_1 = 6$, $y_1 = 8$

We know, $m = \frac{\Delta y}{\Delta x} = \frac{y_1 - y_0}{x_1 - x_0} = \frac{8-6}{3-2} = 1.25 > 1$

here, $m > 1$

So, $x_{new} = \left(x_{old} + \frac{1}{m}\right)$; x value will round-up

$y_{new} = y_{old} + 1$

So,

x_{old}	y_{old}	x_{new}	y_{new}	x plot	y plot
2	3	$\left(2+\frac{1}{1.25}\right) = 2.8$	$(3+1) = 4$	3	4
2.8	4	$\left(2.8 + \frac{1}{1.25}\right) = 3.6$	$4+1 = 5$	4	5
3.6	5	$3.6 + \frac{1}{1.25} = 4.4$	$5+1 = 6$	4	6
4.4	6	$4.4 + \frac{1}{1.25} = 5.2$	$6+1 = 7$	5	7
5.2	7	$5.2 + \frac{1}{1.25} = 6$	$7+1 = 8$	6	8

So plotting point, $(x, y) = \{(3,4), (4,5), (4,6), (5,7), (6,8)\}$

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Problem2: Use the **Digital Differential Analyzer (DDA) algorithm** to find and plot the intermediate points for the line segment between **(2,3) & (8,6)**.

➔ **Is given,**

$X_1 = 2$ $\quad Y_1 = 3$

$X_2 = 8$ $\quad Y_2 = 6$

We Know,

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{6-3}{8-2} = 0.5 < 1$$

if $m < 1$ then,

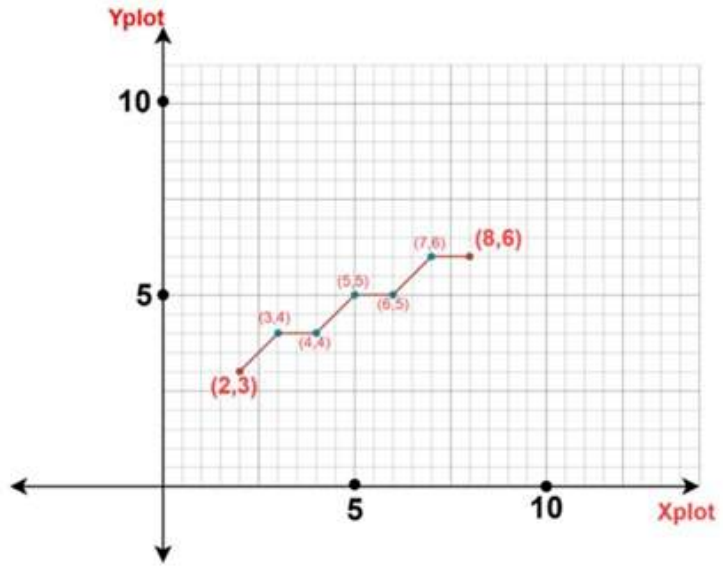
- $x_{new} = x_{old} + 1$
- $y_{new} = round(y_{old} + m)$

x_{old}	y_{old}	x_{new}	y_{new}	x_{plot}	y_{plot}
2	**3**	$(x_{old} + 1) = (2+1) = 3$	$(y_{old} + m) = 3 + 0.5 = 3.5$	3	$round(3.5) = 4$
3	3.5	$(x_{old} + 1) = (3+1) = 4$	$(y_{old} + m) = 3.5 + 0.5 = 4$	4	$round(4) = 4$
4	4	5	4.5	5	5
5	4.5	6	5	6	5
6	5	7	5.5	7	6
7	5.5	8	6	**8**	**6**

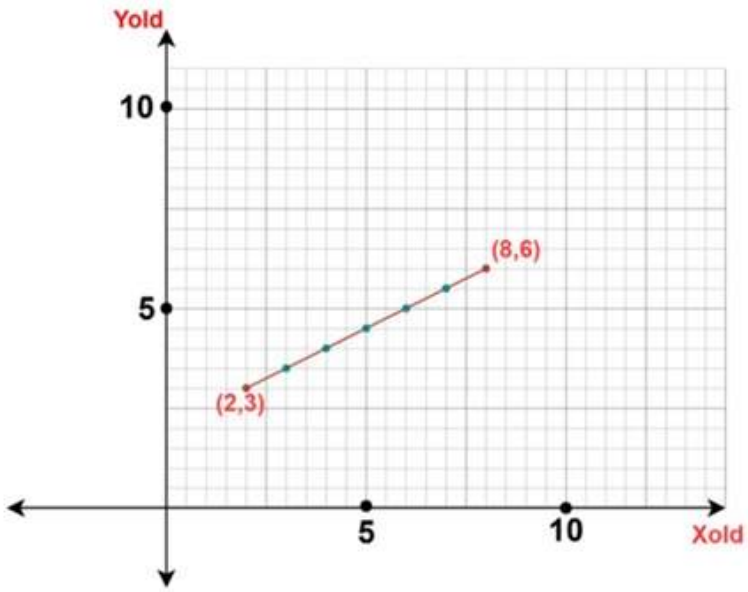
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- $y_{new} = round(y_{old} + m)$

x_{old}	y_{old}	x_{new}	y_{new}	x_{plot}	y_{plot}
2	**3**	$(x_{old} + 1) = (2+1) = 3$	$(y_{old} + m) = 3 + 0.5 = 3.5$	3	$round(3.5) = 4$
3	3.5	$(x_{old} + 1) = (3+1) = 4$	$(y_{old} + m) = 3.5 + 0.5 = 4$	4	$round(4) = 4$
4	4	5	4.5	5	5
5	4.5	6	5	6	5
6	5	7	5.5	7	6
7	5.5	8	6	**8**	**6**



We See the **Staircase (Jagged)** Effect



The actual straight line is smooth, but due to pixel approximation, it appears jagged on a digital display

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Problem3: Use the **Digital Differential Analyzer (DDA) algorithm** to find and plot the intermediate points for the line segment between **A= (2,2) & B= (2,7)**.

➔ **Is given,**

$X_1 = 2$ $\quad$ $Y_1 = 2$

$X_2 = 2$ $\quad$ $Y_2 = 7$

We Know,

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{7-2}{2-2} = \frac{7-2}{0} = \textit{undefined}$$

So Theoretically,

DDA does not allow vertical lines.
it works with special adjustments to handle them.

Special Adjustments,

A= (2,2), B= (2,7)

$$\gg \Delta x = x_2 - x_1 = 2 - 2 = 0$$

$$\gg \Delta y = y_2 - y_1 = 7 - 2 = 5$$

$$\gg No.\,of\ step = 5 \longrightarrow \text{[largest value between } \Delta x \text{ \& } \Delta y\text{]}$$

$$\gg Step\ size\ for\ x = \frac{\Delta x}{No.\,of\ step} = \frac{0}{5} = 0$$

$$\gg Step\ size\ for\ y = \frac{\Delta y}{No.\,of\ step} = \frac{5}{5} = 1$$

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x_{old}	y_{old}	x_{new}	y_{new}	x_{plot}	y_{plot}
2	**2**	2+0 = 2	2+1 = 3	2	3
2	3	2+0 = 2	3+1 = 4	2	4
2	4	2	5	2	5
2	5	2	6	2	6
2	6	2	7	**2**	**7**

Table: Intermediate Points Calculated Using DDA

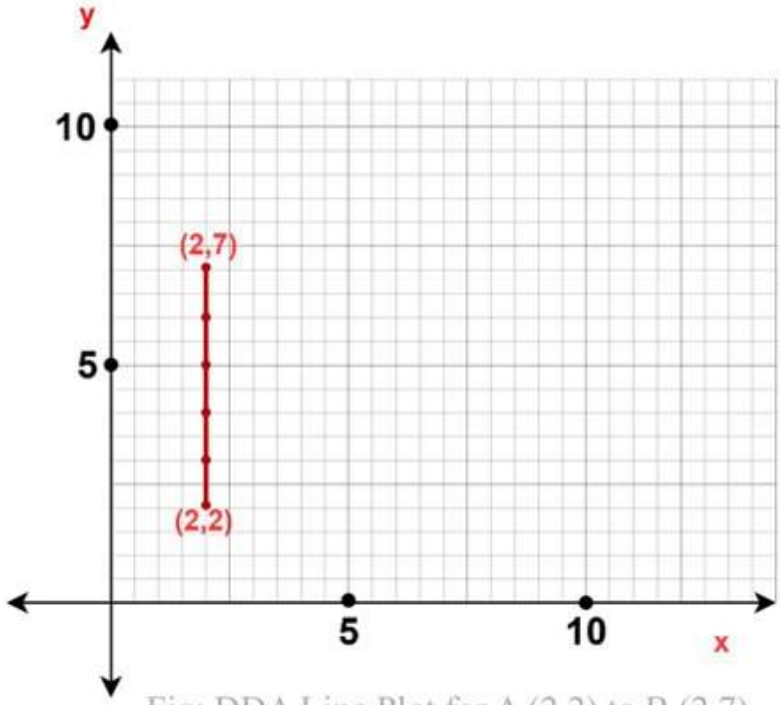


Fig: DDA Line Plot for A (2,2) to B (2,7)

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Ans to the Question no 3 :

মনে রাখবো কখনো, horizontal line, $\Delta y = 0$

vertical line, $\Delta x = 0$

এবং, m এর পরিবর্তে step size দিয়ে কাজ করবো,

for horizontal line,

$\Delta y = 0,$

$y_{new} = y_{old} + \Delta y$

$= y_{old} + 0$

$= y_{old}$

for vertical line,

$\Delta x = 0,$

so, $x_{new} = x_{old}$

$y_{new} = y_{old} + s.s(y)$

$x_{new} = x_{old} + \text{Step size of } (x)$

$$\text{Step size } (x) = \frac{\Delta x}{\text{no. of step}}$$

$$\text{Step size } (y) = \frac{\Delta y}{\text{no of step}}$$

$$\text{no of step} \propto \max(\Delta x, \Delta y)$$

Analytical Math

① use DDA to find the ten points for the line with equation, $7x+5y-40=0$ considering the initial point as $(0,8)$.

Ans: Given line $\Rightarrow 7x+5y-40=0$

Convert to slope form: $5y = 40-7x$

$$\Rightarrow y = \frac{40}{5} - \frac{7}{5}x$$

$$\Rightarrow y = 8 - \frac{7}{5}x$$

So, $m = -\frac{7}{5} = -1.4$.

$|m| = 1.4$

$m > 1$ so, DDA takes y as the driving variable.

steep line

$\Delta y = -1$ [decrease by 1]

$$\Delta x = \frac{\Delta y}{m} = \frac{-1}{-1.4} = .714$$

So, DDA rule:

$$x_{k+1} = x_k + .714$$

$$y_{k+1} = y_k - 1$$

$$\left[m = \frac{\Delta y}{\Delta x} \right]$$

① $x = 0$, $y = 8$ $\quad (0,8)$

② $x = 0 + .714 = .714 = 1$

$y = 8 - 1 = 7$

$(1,7)$

Sub.: .....

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③ $x = .714 + .714 = 1.4 = 1$

$y = x - 1 = 6$ (1, 6)

④ $x = 1.4 + .714 = 2.1 = 2$

$y = 6 - 1 = 5$ (2; 5)

⑤ $x = 2.1 + .714 = 2.8 = 3$

$y = 5 - 1 = 4$ (3, 4)

⑥ $x = 2.8 + .714 = 3.5 = 4$

$y = 4 - 1 = 3$ (4, 3)

⑦ $x = 3.5 + .714 = 4.2 = 4$

$y = 3 - 1 = 2$ (4, 2)

⑧ $x = 4.2 + .714 = 4.9 = 5$

$y = 2 - 1 = 1$ (5, 1)

⑨ $x = 4.9 + .714 = 5.6 = 6$

$y = 1 - 1 = 0$ (6, 0)

⑩ $x = 5.6 + .714 = 6.3 = 6$

$y = 0 - 1 = 1$ (6, -1)

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② Use DDA algorithm to find the first five points for the line : $4x+3y-24=0$, Initial value $(0, 8)$.

Ans: Given: $4x+3y-24=0$

$$\Rightarrow 3y = 24-4x$$

$$\Rightarrow y = \frac{24}{3} - \frac{4}{3}x$$

$$\Rightarrow y = 8 - \frac{4}{3}x$$

So, $m = -\frac{4}{3} = -1.33$ $[|m| = 1.33]$

$\Delta y = -1$, $m = \frac{\Delta y}{\Delta x}$

$$\Rightarrow \Delta x = \frac{\Delta y}{m}$$

$$\Rightarrow \Delta x = \frac{-1}{-1.33}$$

$$\Rightarrow \Delta x = .75$$

So,

② $x = x + .75$

$y = y - 1$

① $x = 0$

$y = 8$ $(0, 8)$

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② $x = 0 + .75 = .75 = 1$

$y = 8 - 1 = 7$ (1,7)

③ $x = .75 + .75 = \cancel{1.75}\ 1.5 = 2$

$y = 7 - 1 = 6$ (2,6)

④ $x = 1.5 + .75 = 2.3 = 2$

$y = 6 - 1 = 5$ (2,5)

⑤ $x = 2.3 + .75 = 3.1 = 3$

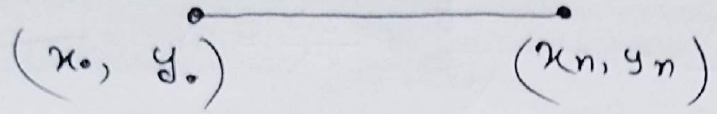
$y = 5 - 1 = 4$ (3,4)

Topic: Derivation

DDA

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Derivation ⟹ DDA



We know,

$$m = \frac{\Delta y}{\Delta x} = \frac{y_n - y_0}{x_n - x_0}$$ and, $y = mx + c$

Let, $m < 1$,

now, Initial pixel, $x_i = x_0$

$y_i = y_0$

next pixel, $x_{i+1} = x_i + \Delta x$

~~$y_{i+1} = y_i + \Delta y$~~ $y_{i+1} = m x_{i+1} + c$

now, we know,

(আমরা বলতে পারি $y_i = (x_i)m + c$)

$$y_{i+1} = m x_{i+1} + c$$

$$= m(x_i + \Delta x) + c$$

$$= m x_i + m \Delta x + c$$

$$= (m x_i + c) + m \Delta x$$

$$= y_i + m \Delta x$$

$$= y_i + m$$

[$\because$ here $\Delta x = 1$]

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So, when, $m < 1$,

next pixel, $\{ x_{i+1}, \text{round}(y_{i+1}) \}$

where, $x_{i+1} = (x_i) + 1$

$y_{i+1} = (y_i) + m$

when $m > 1$,

next pixel $\{ \text{round}(x_{i+1}), y_{i+1} \}$

where, $x_{i+1} = (x_i) + \frac{1}{m}$

$y_{i+1} = (y_i) + 1$

when, $m = 1$,

next pixel $\{ x_{i+1}, y_{i+1} \}$

where $x_{i+1} = (x_i) + 1$

$y_{i+1} = (y_i) + 1$

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